Grey wolf optimization

Pseudo code

Start

Input objective function, n, max iterations
initialize n grey wolves randomly
Evaluate fitness of all wolves

Identify Alpha, beta & delta
(Alpha = best solution)
(Beta = second best solution)
(Delta = third best solu
for t = 1 to max_iterations
update parameter a = 2 - 2*(t)/max_iterations
for each wolf i
Generate values r1, r2

calculate A1 = 2 * a * r1 - a
calculate C1 = 2 * r2

calculate D_alpha = |C1 * Alpha - X(i)|
calculate X1 = Alpha - A1 * D_alpha

Generate random values r1, r2

calculate A2 = 2 * a * r1 - a
calculate C2 = 2 * r2

calculate D_beta = |C2 * beta - X(i)|
calculate X2 = beta - A2 * D_beta

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Generate random values $r_1$, $r_2$

Calculate $A_3 = 2 + \alpha + r_1 - \alpha$
Calculate $C_3 = 2 * r_2$

Calculate $D_{delta} = |C_3 * delta - x(i)|$
Calculate $x_3 = Delta - A_3 * delta$

Calculate position of wolf:

$$x(i) = (x_1 + x_2 + x_3)/3$$

End for

Evaluate fitness for all search agents update alpha, beta & delta values

End for

Output alpha & its fitness

End

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# Grey wolf optimizer & smart grid energy management

The paper focuses on reducing electricity cost & peak power demand in smart grids using the grey wolf optimization algo. It considers electricity consumption, solar photovoltaic energy & time based electricity pricing. The algorithm helps schedule electrical appliances during suitable time period to improve energy efficiency & reduce electricity bills.

## Code Summary

The python implementation creates a small smart grid environment with six time slots, 5 electrical appliances, electricity prices & solar energy availability. It applies the grey wolf optimization algo to find suitable appliance operating times. The fitness function calculates electricity cost & peak demand & the algorithm selects the schedule with the lowest fitness value

## Output Summary

The electricity cost decreases from 1906 to 143 units, while peak demand decreases from 11kW to 5kW. This represents a 92.7% reduction in electricity cost & a 54.55% reduction in peak de